

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

CE-701 (GS)

B.E. VII Semester Examination, June 2020

Grading System (GS)

Design of Hydraulic Structure

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. What are the principal components of a hydroelectric scheme? Discuss the utility of each component.
2. Discuss briefly the various types of energy dissipaters used under different relative positions of T.W.C. and J.H.C.
3. a) What is meant by reservoir? What is its use?
b) What are the causes of failure of earth dam?
4. Describe the brief various investigations required for reservoir planning.
5. How will you calculate the capacity of reservoir by mass inflow and mass demand curves?
6. Discuss practical profile of gravity dam.
7. What is meant by energy dissipaters? Explain energy dissipaters for various cases with basic principle of energy dissipater.

OR

Design the floor on Bligh's theory using coefficient of creep=10. Sketch the longitudinal section of fall. Check the design by Khosla's theory and make changes if necessary. Safe exit gradient may be taken equation to $1/5$.

8. Write short notes on the following.(any two)
 - a) Rock fill Dams
 - b) Life of Reservoir
 - c) Weir and Barrage
 - d) Purpose of Galleries
